DELTA CUP 2026 · RACKBOTGEN3

RBG3-MOTION v1 — Schematic Review

What to fix, what to flag, what to improve before the JLCPCB order

Source: `rbg3-motion.kicad_sch` (netlist from the PCB, parity-clean) · spec v0.1.19 · 2026-10-21

Summary

Netlist checked pin by pin against spec 02 §3: every GPIO, the UART addresses (MS1/MS2), the CLK tie, the motor pin order and the DevKitC-1 header pinout match. No wiring errors found. The open items are missing parts and system-level risks.

3	5	6	✓
Fix before order (cheap parts, footprints only)	Flag (verify on the bench or decide)	Improve (v1 nice-to-have or v2)	Pin map, addresses, dividers, UART verified

FIX Before ordering

ISSUE	WHY IT MATTERS	FIX
+6V_SERVO has no capacitors (net = J3, J14, J15 only)	MG90S start/stall pulls ≈ 0.7–1 A each in spikes. The buck is off-board on a lead, so the rail sags at the servo and the servos jitter or brown out.	Add 220–470 µF / 16 V electrolytic + 100 nF at J14/J15.
Q1 base has no pull-down (GPIO48 → R24 1 k → Q1)	GPIO48 floats during reset and boot, so the buzzer can chirp or half-conduct. GPIO48 also drives the onboard RGB LED on some DevKits.	Add 10 k–47 k from BUZ_B to GND.
No local bulk on +5V / +3V3	+3V3 leaves the board on 7 connectors (J8–J13, J16) and feeds 4 TMC VIO pins with only the DevKit LDO's caps. +5V (WS2812, buzzer) has none.	Add 10 µF + 100 nF on +3V3 near the sockets and 10 µF on +5V at J2.

FLAG Verify or decide

ITEM	RISK	ACTION
VMOT inrush (870 µF total)	SPICE shows 120–220 A for ≈ 100 µs on every E-stop release. That can trip the BMS short-circuit protection (which also kills the Pi) and wears the relay contact.	Bench test E-stop release ×20. If the BMS trips, add an NTC or soft-start, or reduce the bulk to ≈ 470 µF.
E-stop detection 370 ms with drivers disabled	VMOT_SENSE decays slowly through the 118 k divider. That misses the 50 ms target.	Wire a 2nd NC contact of the mushroom to **J13 AUX_IN**. Instant, no PCB change.
Ground loop via USB	Motor return current can share the Pi → USB → ESP32 → board GND path with J1–, which causes noise and USB drop-outs.	Return the 5 V and servo buck grounds at J2/J3, not at the battery. Keep the J1 GND lead short and thick.
WS2812 at 3.3 V	Out of spec (VIH 3.5 V). It works on most parts.	Use an SK6812 / WS2812B-V5, or see the buffer item on the next slide.
J2 / J3 not reverse-protected	A swapped screw-terminal lead kills the servos or the LED.	Use keyed connectors (JST-VH / XT30), or polarity marks plus a meter check at bring-up.

IMPROVE

Worth adding

IMPROVEMENT	BENEFIT
74AHCT1G125 footprint on LED_DIN (0 Ω bypass)	Proper 5 V level for the WS2812. Leave it unfitted unless needed.
Test points: VMOT, +5V, +6V, +3V3, GND, TMC_UART, EN_N	Faster bring-up and probing with modules fitted.
TVS on +6V_SERVO / +5V (SMAJ6.0A)	If an off-board buck fails short high-side, 16.8 V reaches the servos and LED. The TVS crowbars it instead.
Route DIAG pins (driver side pins) to spare GPIO	StallGuard jam detection on the Y extractor, sensorless homing as a backup.
3V3 on switch headers J8–J13	Unused today. A harness short there browns out the ESP32. Add a 100 Ω series resistor or leave it unpopulated.
Reverse-polarity P-FET on VMOT (v2)	Already noted in the spec. Removes the "fault current through USB GND" path.

Schematic readability

- **Title block empty**: fill in Title, Rev (v1 / spec v0.1.19), Date, Company.
- **Overlapping text** at the J1–J3 values, D1/C9, the R1/R2 and R3/R4 dividers, J18/J19 and the C10/C11 labels. Hard to read in the PDF.
- **Label-only connectivity**: no drawn wires, so the power flow is invisible. Group by function with boxed sections (Power, Drivers, MCU, I/O, Sense).
- **Divider diagonal wires** at R1/R2 and R3/R4 cross the labels.
- Add **text notes** on the sheet: UART addresses, NC home switches, "J18 pin 2 NC on purpose", E-stop is upstream.
- Add a **power-tree note**: VMOT → drivers, ext. bucks → J2/J3, USB → DevKit 3V3.
- Show **pin 1 / polarity** on J2/J3 symbols to match the silkscreen.
- Optional: split into 2–3 hierarchical sheets (A3 is crowded).

OK Verified, no change

- DevKitC-1 pin map (J1 1–22, J3 23–44) matches the Espressif layout.
- All GPIOs match spec §3.1. Strapping pins (0/3/45/46), USB (19/20), UART0 and PSRAM (33–37) are left unconnected.
- ADC on GPIO1/2 = ADC1, so it works with Wi-Fi on.
- MS1/MS2 give addresses 0/1/2/3. CLK is on GND (internal clock).
- EN_N has a 10 k pull-up, so the drivers stay off during reset and when USB is lost.
- UART: TX through 1 k, RX direct, pin 4 + JP pin 5. Sim margins are fine.
- Dividers: 2.56 V at 16.8 V, 2.18 V at 24 V. A fault current at 29 V is limited to < 0.3 mA by 100 k.
- D2 flyback is the right way round (K at +5V). Q1 base drive ≈ 2.6 mA is enough.
- Switch inputs: pull-up + 1 k + 100 nF ≈ 1 ms RC, fine for debounce.
- SMBJ18A: standoff 18 V > 16.8 V. Sim peak 17.6 V.

Action list

#	ACTION	TYPE	COST
1	+6V_SERVO bulk 220–470 µF + 100 nF at J14/J15	FIX	2 parts
2	BUZ_B 10 k pull-down	FIX	1 × 0805
3	10 µF + 100 nF on +3V3; 10 µF on +5V	FIX	3 × 0805
4	Fill in the title block and fix overlapping text	IMPROVE	10 min
5	E-stop 2nd contact → J13 AUX_IN	FLAG	wiring only
6	Bench test the inrush and BMS on E-stop release	FLAG	bring-up
7	Test points, LED buffer footprint, DIAG routing	IMPROVE	optional

Items 1–3 add new purchased parts, so update `docs/parts.csv`, Notion and the budget (CLAUDE.md rule).